

Standardizing RNA-seq Analysis of Fungal Pathogens Using BRC-Analytics and Agentic AI

A Candidozyma auris Case Study

Nekrutenko et al.

The *C. auris* Threat

- First isolated 2009 (Japan), now global spread
- **CDC urgent threat** — first fungal pathogen with this designation
- Multidrug resistance (often to ALL major antifungal classes)
- **30-60% mortality rates**
- Persists on surfaces, forms biofilms on medical devices
- WHO critical-priority pathogen

C. auris Sequencing Landscape

Assay	BioProjects	Runs
WGS	168	26,201
RNA-seq	64	812

- WGS: outbreak surveillance (avg 156 runs/project)
- RNA-seq: academic research (avg 13 runs/project)
- **27% of projects are RNA-seq — standardization is priority**

The Methodology Problem

Survey of 20 published RNA-seq studies:

Category	Finding
Reference Genome	60% use B8441, but versions vary
Tools	HISAT2 (7), STAR (5), DESeq2 (12)
Version reporting	Often incomplete or absent

Result: Reproducibility challenges, cross-study comparison difficult

Our Solution

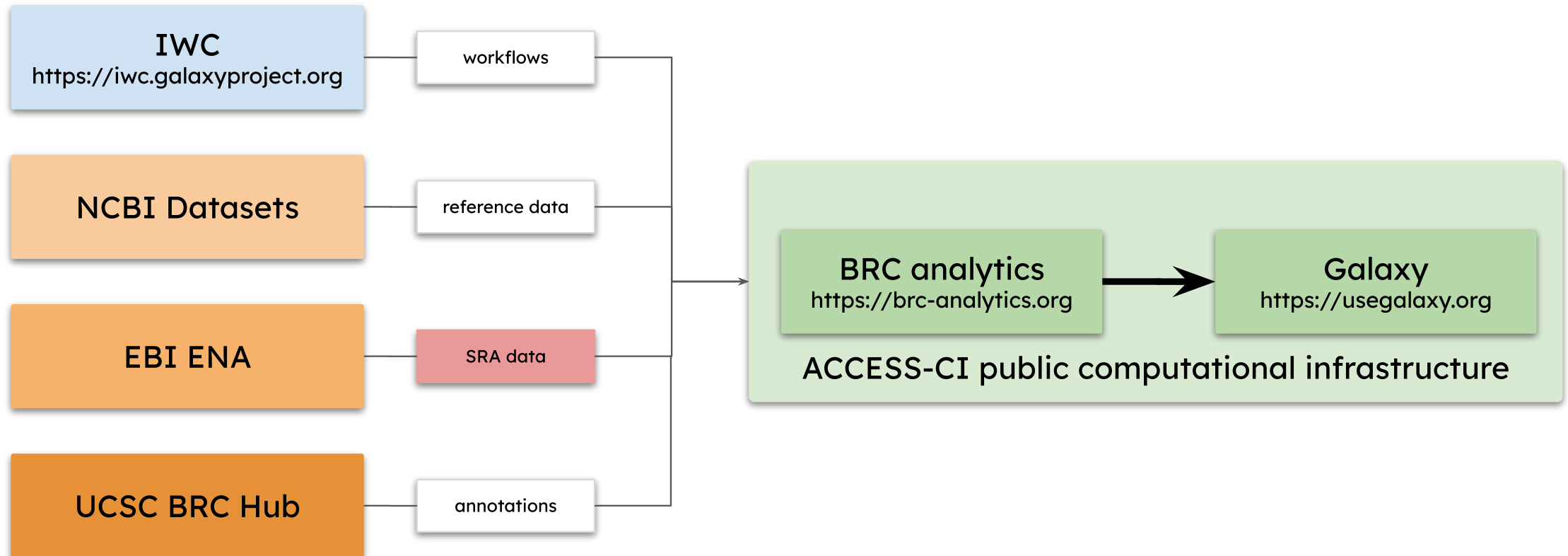
BRC-Analytics + Galaxy + Agentic AI

- Standardized workflows with **locked versions**
- **Automatic provenance** tracking
- Natural language interaction via **Claude Code Agent**

BRC-Analytics Platform

- **Browser-based** — no local installation
- NIAID-funded Bioinformatics Resource Centers
- Data sources:
 - NCBI Datasets (5,060 assemblies)
 - UCSC Genome Browser
 - EBI ENA
- **IWC community-curated workflows**
- ACCESS-CI infrastructure at TACC

BRC-Analytics Platform



Two Validation Studies

Study	Key Finding	BioProject
Santana 2023 (<i>Science</i>)	SCF1 adhesin for biofilm/virulence	PRJNA904261
Wang 2024 (<i>Nat Comm</i>)	Glycan-lectin in colonization	PRJNA1086003

Both: B8441 reference, modern tools, adhesion phenotypes

Why Agentic AI?

Two aspects of biological data analysis **resist automation**:

1. **SRA ↔ Manuscript mapping**

Sample groupings, replicates rarely obvious from metadata

2. **Cross-version comparison**

Different genome annotations = different gene IDs

Standard workflows can't solve these

Why NOT Web-Based LLMs?

ChatGPT, Claude, Gemini in browser

- Artifacts hard to track, version, reproduce

Claude Code, Gemini Code Assist (local)

- All artifacts preserved locally
- Git versioning
- Can interact with Galaxy API

Our Setup

Claude Code Agent (CCA)

- Configured with Galaxy API key
- Runs on local machine
- All interactions recorded

Verification protocol:

1. CCA proposes plan
2. Human reviews/modifies
3. CCA executes

Organizing Data with CCA

Prompt: *"Split Galaxy collection #244 into conditions from manuscript..."*

CCA:

- Downloads SRA metadata
- Reads manuscript PDFs
- Maps accessions → experimental conditions
- Creates labeled collections in Galaxy

The Annotation Challenge

Version	Gene ID format
Published studies	B9J08_ 001458 (6-digit)
Current GCA_002759435.3	B9J08_ 03708 (5-digit)

Solution: NCBI official `old_locus_tag` mapping

- GTF file contains authoritative gene ID correspondence
- Validated by **100% protein sequence identity**

AI Mistake: LFC Correlation

Initially CCA proposed matching genes by similar fold-change values:

- **Appeared successful:** $R^2 = 0.9996$
- **Actually wrong:** Only 1% of matches correct (2/203)

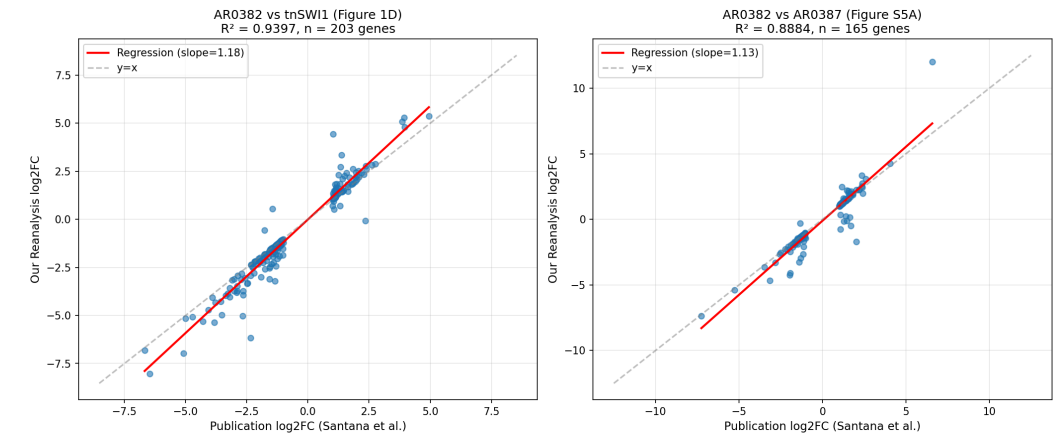
High R^2 was coincidental—genes with similar LFC by chance

Lesson: Validate AI outputs against authoritative sources

Santana et al. Validation

Comparison	Genes	R ²	Direction
tnSWI1 vs AR0382	203	0.94	99%
AR0387 vs AR0382	165	0.89	97%

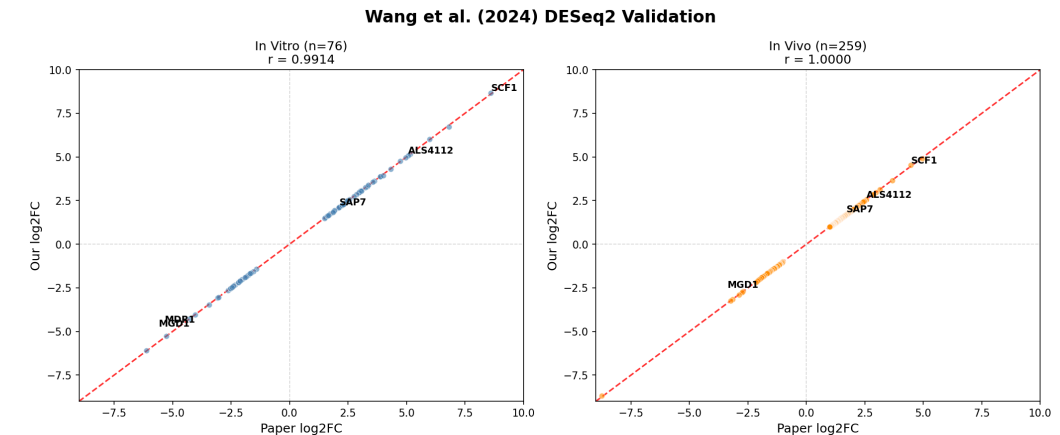
SCF1: -6.68 (pub) vs -6.82 (ours)



Wang et al. Validation

Condition	Genes	R ²	Direction
In vitro	76	0.98	100%
In vivo	259	0.9998	100%

Strong correlation with official mapping



The Provenance Problem

AI agents can bypass Galaxy's tracking:

- Direct API manipulation → loses reproducibility
- External scripts → not in Galaxy history

Our solution:

- Constrain CCA to use **Galaxy tools via API**
- Custom code → **JupyterLite notebooks** (AI-free)

Key Results

Metric	Result
Santana correlation	$R^2 = 0.89-0.94$
Wang correlation	$R^2 = 0.98-0.9998$
SCF1 finding	Confirmed
Official NCBI mapping	Works across annotation versions
Provenance	Maintained via Galaxy + notebooks

LLMs for Reproducible Research

Do:

- Use **agentic tools** (Claude Code, Gemini Code Assist)
- Automatic artifact tracking
- Version-controlled repositories

Don't:

- Use chat interfaces for research
- Browser-based Claude/ChatGPT lose provenance

Why Galaxy for AI Integration?

Feature	Galaxy	Nextflow
Tool metadata	Structured XML	Parse docs/source
API	Stateful	Log parsing
HPC access	ACCESS-CI (free)	Configure manually
Interface	Browser-based	Command-line

AI agents work better with structured APIs

Future Vision

Current: Manual coordination between BRC-Analytics, Galaxy, CCA

Future: AI embedded in Galaxy

- Natural language → workflow selection
- Automatic parameter configuration
- Result interpretation
- Full provenance maintained

Conclusions

1. **BRC-Analytics** enables standardized pathogen genomics
2. **Agentic AI** handles non-automatable tasks
3. **Local AI tools** (not web chat) preserve provenance
4. **Galaxy's structured API** enables AI integration
5. **First integration** of data repos + workflows + agentic AI

Thank You

Resources:

- BRC-Analytics: <https://brc-analytics.org>
- Galaxy histories: usegalaxy.org
- Code: github.com/nekrut/claude-projects

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